

Security Evaluation of a Face Recognition System

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Abstract This project investigates the vulnerability of face recognition models to adversarial examples by analyzing their classification performance under controlled perturbations and evaluating possible countermeasures. A dedicated test set was constructed by selecting 100 identities from the **VGGFace2 dataset**, each represented by ten images, and experiments were carried out using the **Inception Resnet V1** model pre-trained on the same dataset. After establishing baseline performance on clean inputs, a variety of adversarial attacks were applied to assess their impact on accuracy. To examine the transferability of these examples, a second model based on the **ResNet50 architecture** was evaluated using adversarial inputs originally crafted to fool the first model. Finally, some defense mechanisms were implemented and tested to measure their effectiveness in mitigating the performance degradation caused by such attacks.

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1 Introduction

Face recognition technologies have become increasingly widespread in recent years, finding application in areas ranging from personal device authentication to large-scale identity management systems. Their adoption is largely due to the progress achieved by deep learning models, which have demonstrated strong generalization capabilities and high classification accuracy on large and varied datasets. However, alongside these advances, a number of critical vulnerabilities have emerged, particularly concerning the susceptibility of such models to adversarial examples.

Adversarial examples are inputs that have been intentionally perturbed in a way that is imperceptible to the human eye, but which can cause machine learning models

to produce incorrect or even highly misleading outputs. In the context of face recognition, this vulnerability raises concerns about the reliability and security of systems deployed in practical scenarios. Understanding how these attacks affect model performance, and exploring strategies to mitigate their effects, has therefore become an important area of research.

This project focuses on evaluating the robustness of a face recognition model when subjected to a range of adversarial attacks. The analysis is structured around a classification task based on a subset of the **VGGFace2 dataset**, from which 100 identities and a fixed number of images per identity were selected to form a custom test set. The main model under investigation is **Inception Resnet V1**, pre-trained on the full **VGGFace2 dataset** and used to classify input images based on learned facial representations.

Once the model's baseline performance is measured on clean images, a set of adversarial attacks is applied. These include both single-step and iterative methods, evaluated in their untargeted and targeted variants, where supported. To assess the transferability of adversarial examples specifically crafted against the first model, a second model based on the **ResNet50 architecture** is evaluated using those same inputs. Finally, the project introduces some defense mechanisms, whose effectiveness is evaluated in terms of their ability to reduce the impact of these attacks on classification accuracy.

The overall goal is to obtain a clearer understanding of how different models behave under adversarial conditions, how attacks differ in their effectiveness and transferability, and to what extent defenses can help restore reliability in the presence of adversarial inputs.

2 | Test Set

The experimental analysis begins with the construction of a dedicated test set derived from the **VGGFace2 dataset**, which contains over 3.3 million images of more than 9,000 identities, offering significant variability in terms of pose, age, illumination, and facial expression. For the purposes of this study, 100 identities were selected, each associated with 10 image samples.

To support this process, the dataset was accompanied by the `identity_meta.csv` file, which includes metadata for each identity such as ID, name, and gender. Identities selection could be performed either manually or randomly using a dedicated script. In the random case, the metadata was first split into two groups based on gender. The objective was to collect 50 valid identities from each group by iterating over the corresponding entries. For each candidate, the script verified the existence of the associated folder in the dataset and checked whether it contained at least 10 images. Only identities satisfying both conditions were considered.

If fewer than 50 valid identities were found in one group, the remaining identities were drawn from the

other one, depending on how many were still needed and how many unused entries remained. This procedure ensured that the test set included as close as possible to 100 valid identities, while respecting the availability constraints of the original dataset.

Once the full list of valid identities was obtained, 10 images were extracted for each identity and copied into a new directory dedicated to the test set, with each identity assigned to its own subfolder named accordingly. The list of selected identities was recorded in a file named `test_set.csv`, which contains the corresponding metadata rows. When the selection is performed manually, this file must be created in advance by the user, after which the image extraction process follows the same logic used in the random case. Finally, the test set was alphabetically ordered by identity name to maintain consistency in downstream processing.

3 | Evaluation on NN1 (Inception Resnet V1)

3.1 | Model Overview

Following the construction of the test set, the objective was to assess the performance of a face recognition model in a classification setting. For this purpose, we employed the **Inception Resnet V1** model provided by the `facenet-pytorch` library. This model, which will be referred to as NN1 throughout the project, is designed to take as input RGB face images of size 160×160 pixels and to output a 512-dimensional embedding vector that encodes the identity-related features of the face.

The embeddings produced by NN1 are designed to be robust to variations in pose, lighting conditions, and facial expressions, making them suitable for real-world face recognition scenarios. However, in order to ensure optimal performance, input images must be properly preprocessed, particularly through face alignment and cropping, which help to standardize the facial region across samples. To handle this task, we

employed the **MTCNN (Multi-task Cascaded Convolutional Networks)** model, which is also provided by the `facenet-pytorch` library and is commonly used in combination with NN1. **MTCNN** detects the facial region within each image and returns a cropped and aligned version centered on the detected face, resized to the required input resolution. This preprocessing step helps standardize facial geometry across the dataset and is essential for maintaining the discriminative quality of the embeddings.

Since our goal is to assign each input image to one of the known identities, we take advantage of a feature offered by `facenet-pytorch` that allows us to attach a linear classification layer to NN1. This layer maps the 512-dimensional embeddings to a set of logits, which can then be used to associate each input with the most likely identity by selecting the one with the highest score. In this way, NN1 functions not only as an embedding extractor but also as a complete image classifier.

The version of NN1 used in this project is the one pre-trained on the **VGGFace2 dataset**.

3.2 | Baseline Accuracy on Clean Data

The classification performance of NN1 was evaluated on the clean test set, which consists of 1000 images evenly distributed across 100 identities. Given the classification setting, where each image must be assigned to exactly one identity, we used top-1 accuracy as the evaluation metric, measuring how often the highest-scoring prediction matched the ground truth label.

NN1 achieved an accuracy of 98.60%, correctly classifying 986 out of 1000 test images. Due to the large number of classes, a confusion matrix was not included, as it would offer limited readability in this context. Instead, we analyzed the individual misclassifications. Most of them occurred only once and were spread across different identities. One exception was the identity *Magdalena Gessler*, which was misclassified twice, once as *Emel Sayin*

Metric	Value
Total test samples	1000
Correctly classified	986
Misclassified	14
Accuracy on clean data (NN1)	98.60%

Table 1: Classification performance of NN1 on the clean test set.

and once as *Magdalena Brzeska*. This may indicate higher similarity between those embeddings, potentially due to facial resemblance or stylistic features shared in the images.

Overall, the low number of misclassifications and the even distribution of errors confirm that NN1 performs robustly under clean, unperturbed conditions.

3.3 | Robustness to Adversarial Attacks

3.3.1 | Attack Setup

Following the evaluation of NN1 on clean data, a series of adversarial attacks were applied in order to assess the model's robustness under different perturbation strategies. The attacks were generated using the **Adversarial Robustness Toolbox (ART)**, with NN1 serving as the target model.

Specifically, both error-generic and error-specific variants were implemented for the FGSM, BIM, PGD, and Carlini & Wagner (L_∞) methods. In the case of DeepFool, only the error-generic version was considered, as no error-specific variant is available. For each attack, classification accuracy was measured in terms of top-1 accuracy. In the case of error-specific attacks, we additionally measured the success rate, defined as the proportion of adversarial samples that were classified as the intended target identity.

In the targeted setting, adversarial examples were generated to induce the model to classify each input as a specific, fixed target class. Since the effectiveness of targeted

attacks is highly dependent on the geometry of the decision boundaries and the position of the target class in the feature space, several candidate classes were initially tested for each attack type. For every attack, the target class that produced the highest overall success rate, that is, the greatest number of adversarial samples misclassified as the intended target, was selected for the final evaluation. Notably, the chosen class is always one of the 100 in the balanced test set, each represented by only 10 images. This corresponds to a baseline classification probability of 1% in clean conditions, making any substantial increase under perturbation a clear indication of targeted success rate.

Where computational resources allowed, Security Evaluation Curves (SECs) were produced to analyze the variation in model performance as a function of attack strength. These curves provide a more detailed view of how the model degrades under increasing perturbation intensity.

Finally, for all attacks, including those used to generate the SECs, the maximum perturbation was recorded in terms of the ℓ_∞ norm. This was done to ensure compliance with the constraint specified in the project guidelines, which requires the attack strength to remain within 5% of the input value range. Since our images are normalized in the $[0, 1]$ range, this corresponds to a maximum ℓ_∞ value of 0.05.

3.3.2 | Fast Gradient Sign Method (FGSM)

The Fast Gradient Sign Method (FGSM) was applied to NN1 in both its error-generic and error-specific variants. In each case, adversarial examples were generated by progressively increasing the perturbation magnitude, with ϵ values ranging from 0.010 to 0.100. Only values of ϵ that resulted in perturbations within the allowed $\ell_\infty \leq 0.05$ constraint were considered for evaluation; larger values were deliberately excluded to ensure compliance with the project requirements. The model's

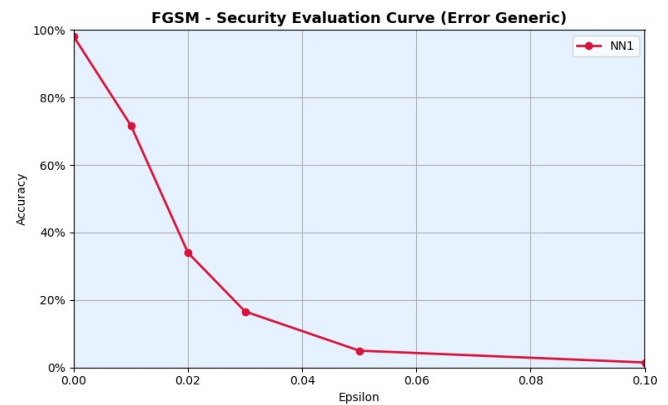


Figure 1: Security Evaluation Curve for FGSM (error-generic variant) applied to NN1.

performance under attack was measured and visualized using Security Evaluation Curves (SEC), allowing a direct comparison between the two attack settings.

Error-Generic Variant. The error-generic version of FGSM aims to induce misclassification without targeting a specific class. As shown in Figure 1, NN1 exhibited a rapid degradation in performance even for low values of ϵ . Starting from the clean accuracy of 98.6%, accuracy dropped to 71.7% at $\epsilon = 0.010$, and continued to decline sharply with increasing perturbation. At $\epsilon = 0.050$, accuracy was reduced to 5%, and at $\epsilon = 0.100$ it reached a minimum of 1.5%. These results underscore the vulnerability of NN1 to simple single-step attacks, which can produce substantial degradation even with limited perturbation strength.

Error-Specific Variant. For the error-specific variant of FGSM, the evaluation focused on how effectively the attack could redirect predictions toward the selected target class. Figure 2 shows the variation of both general classification accuracy and attack success rate across different ϵ values. At $\epsilon = 0.010$, the model retained high overall accuracy (91.3%) and only 1.5% of the inputs were redirected to the target. As ϵ increased, the attack became more effective: the success rate reached 17.1% at $\epsilon = 0.030$, and peaked at 47.1% for $\epsilon = 0.100$, while

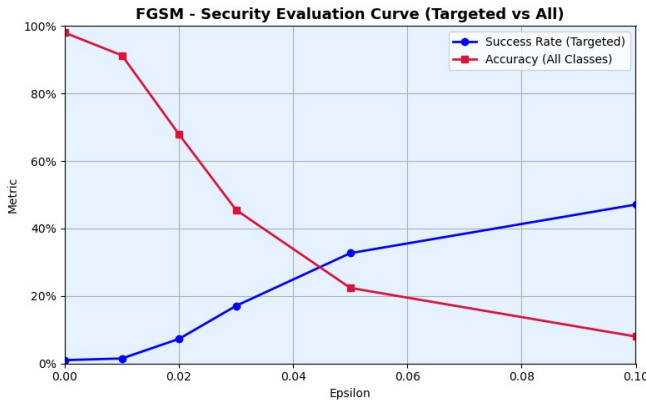


Figure 2: Security Evaluation Curve for FGSM (error-specific variant) applied to NN1.

the general accuracy simultaneously dropped to 8.0%. This divergence illustrates the progressive convergence of predictions toward the intended class. Despite this trend, the overall success rate remained below 50%, confirming that single-step targeted attacks like FGSM are limited in their ability to enforce specific outputs, particularly in high-dimensional multiclass classification problems and when subjected to perturbation constraints.

3.3.3 Basic Iterative Method (BIM)

The Basic Iterative Method (BIM) extends the FGSM approach by applying it multiple times with smaller step sizes, iteratively updating the perturbation while keeping it within a bounded ℓ_∞ norm. In this evaluation, BIM was tested on NN1 using both the error-generic and error-specific variants. Three main parameters were varied independently to observe their impact on model performance: the overall perturbation bound ε , the per-step update magnitude ε_{step} , and the number of iterations.

Among the tested configurations, only those that produced adversarial perturbations within the allowed $\ell_\infty \leq 0.05$ threshold were retained for analysis, while more aggressive settings were excluded to comply with project constraints. Security Evaluation Curves (SEC) were used to visualize the effect of these parameters on

the model's top-1 accuracy and, for the targeted setting, on the attack success rate.

Error-Generic Variant. In the untargeted setting, BIM proved to be highly effective in reducing NN1's classification accuracy. As shown in Figure 3, increasing the perturbation budget ε led to a rapid degradation in performance: accuracy dropped from 98.6% on clean data to 41.8% at $\varepsilon = 0.010$, and then plunged to just 0.5% at $\varepsilon = 0.030$. Beyond that point, no further impact was observed, with accuracy remaining constant at 0.5%, indicating saturation of the attack's effect given the chosen step size and number of iterations.

A similar trend emerged when varying the step size ε_{step} while keeping $\varepsilon = 0.030$ fixed. Very small values such as 0.001 had limited effect, with accuracy at 94.2%. However, even a moderate increase to 0.005 drastically dropped accuracy to 23.3%, and at 0.010 the model was effectively neutralized with just 0.5% accuracy. Raising the step size further to 0.030 produced only a marginal difference (0.1%).

As for the number of iterations, increasing from three to ten resulted in a progressive reduction of accuracy from 0.5% to 0%, with no improvement beyond ten iterations. This indicates that for the fixed budget of $\varepsilon = 0.030$ and $\varepsilon_{step} = 0.010$, most of the attack's strength is already concentrated in the early steps. Overall, these findings demonstrate BIM's capacity to significantly erode classification performance with minimal iteration and perturbation tuning, although its effectiveness tends to saturate quickly once a critical configuration is reached.

Error-Specific Variant. Results for the targeted variant of BIM are shown in Figure 4. At $\varepsilon = 0.010$, the model maintained a high top-1 accuracy of 87.1%, while only 4.9% of the samples were redirected to the target class. A sharp transition was observed at $\varepsilon = 0.030$, where the targeted success rate rose dramatically to 80.1% and remained constant across all higher perturbation values. In contrast, the general accuracy dropped

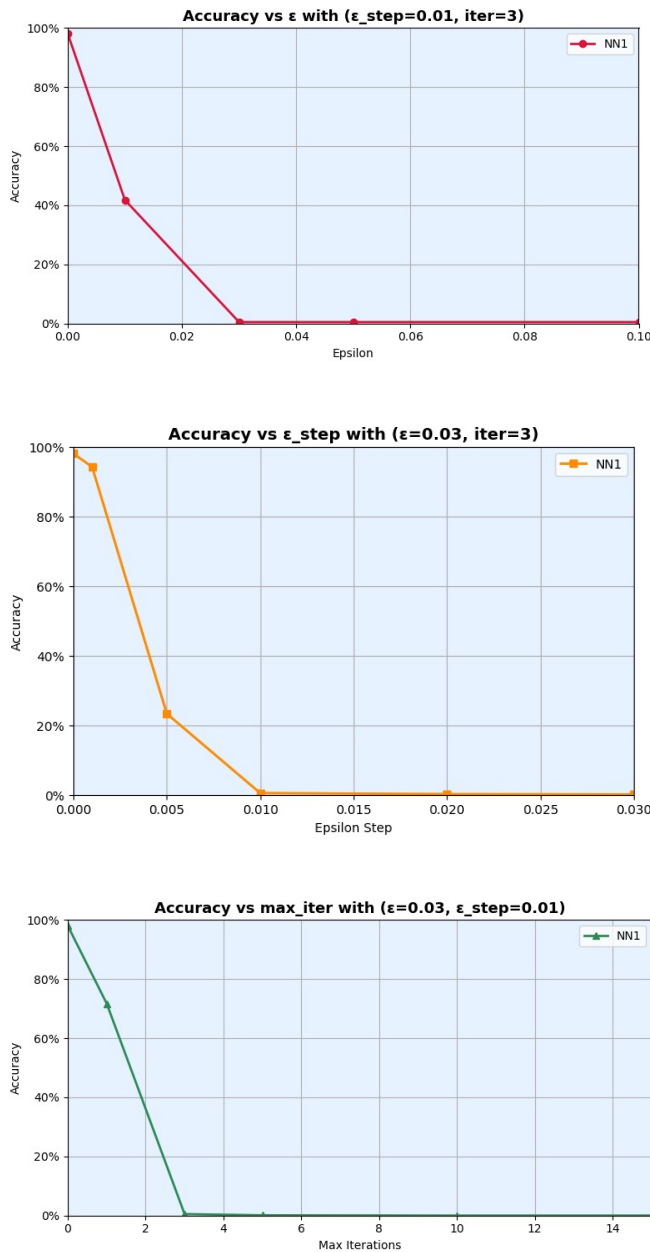


Figure 3: Security Evaluation Curves for BIM (error-generic variant) applied to NN1.

abruptly to 7.5% and likewise stabilized, suggesting that the attack quickly saturated in its effectiveness beyond a certain perturbation threshold.

This saturation trend was mirrored in the variation of ϵ_{step} : increasing the step size from 0.001 to 0.010 produced a dramatic jump in success rate from 1.0% to 80.1%, while a further increase to 0.030 raised the success rate to 91.1%. Meanwhile, the corresponding impact on accuracy was more pronounced, falling from 98.3% to just 3.2%. This suggests that after a critical step size is reached, additional increases primarily serve to further destabilize the model rather than improve targeted control.

A similar effect was observed when increasing the number of iterations: with just three iterations, 79.1% of inputs were successfully redirected, while ten iterations pushed the success rate close to 99.6%, with only marginal gains beyond that. The accuracy, however, dropped steeply in the early stages and plateaued near 1%. This confirms that the majority of redirection takes place within the first few iterations, and that extending the attack mostly solidifies already destabilized predictions.

These results emphasize how BIM, even with relatively mild parameter settings, can achieve high targeted success rates with minimal effort once certain thresholds in ϵ , ϵ_{step} , and number of iterations are surpassed.

3.3.4 | Projected Gradient Descent (PGD)

The Projected Gradient Descent (PGD) method is a widely used first-order adversarial attack that extends the Basic Iterative Method (BIM) by incorporating random initializations. Each input is perturbed within the allowed ℓ_∞ ball by a small random noise, and then iteratively updated through multiple gradient-based steps, with projections ensuring that the perturbation remains within the specified bounds. This random start helps the attack escape local minima and increases its capacity to gener-

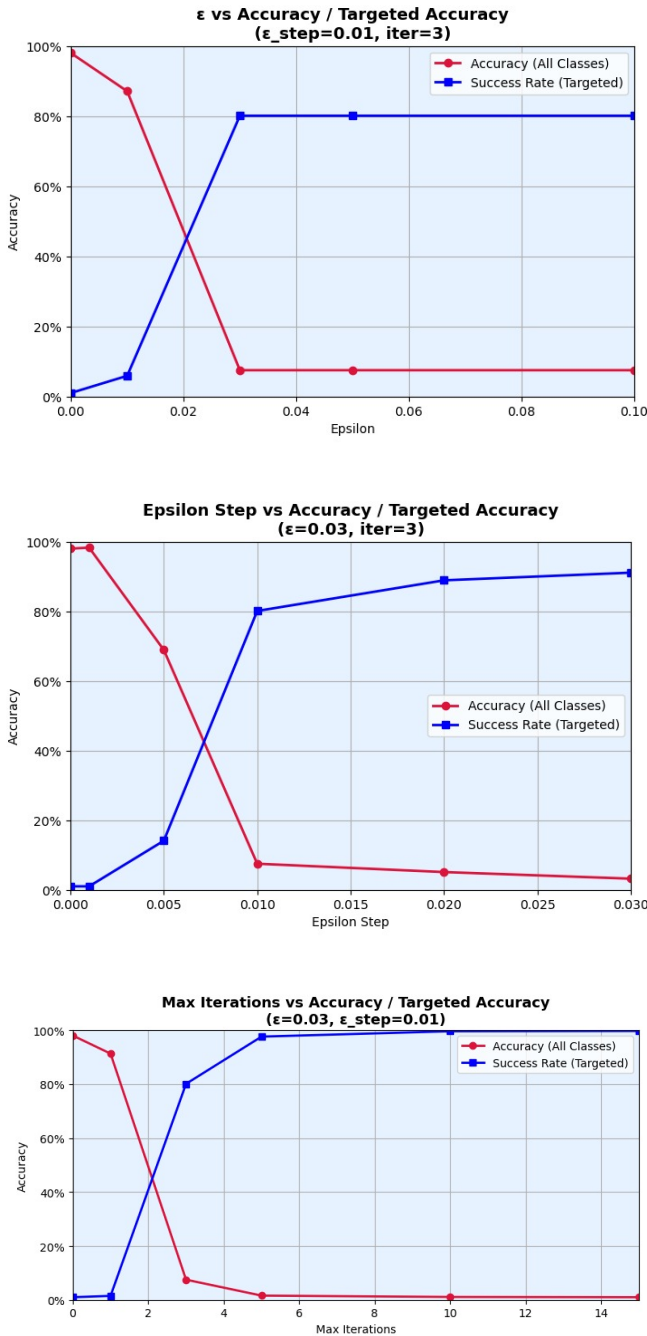


Figure 4: Security Evaluation Curves for BIM (error-specific variant) applied to NN1.

ate stronger adversarial examples.

Both error-generic and error-specific variants of PGD were evaluated by varying key parameters, including the maximum allowed perturbation ϵ , the step size ϵ_{step} , the number of iterations, and the number of random restarts. For each configuration, performance metrics were tracked and visualized using Security Evaluation Curves (SECs), which reflect how the model's robustness degrades as the attack strength increases.

To ensure compliance with the project specifications, only configurations that resulted in adversarial perturbations within the allowed $\ell_\infty \leq 0.05$ threshold were considered. More aggressive settings that would have exceeded this limit were deliberately excluded from the evaluation.

Error-Generic Variant. As shown in Figure 5, PGD proves highly effective in reducing NN1's classification accuracy under the untargeted setting. Already at a relatively low perturbation of $\epsilon = 0.010$, the accuracy drops to 41.4%. Increasing ϵ to 0.030 causes a near-complete collapse of performance, with just 1.4% of inputs correctly classified. Beyond this point, the model's accuracy stabilizes below 1%, reaching as low as 0.3% at $\epsilon = 0.100$.

Similar results are observed when analyzing the effect of the step size ϵ_{step} . An increase from 0.005 to 0.030 sharply reduces accuracy from 27.6% to 0.1%, confirming the attack's steep effectiveness once a sufficient gradient update magnitude is reached. Likewise, increasing the number of iterations consistently amplifies the attack's impact: accuracy drops from 1.4% at three iterations to 0% already by ten. The number of random initializations was also varied, but in this case, no substantial gain was observed, the accuracy remained practically unchanged across one, three, and five restarts, indicating that in this scenario, the attack is already sufficiently strong to saturate performance loss with a single run. Interestingly, this pattern of performance degradation across all tested parameters closely mirrors what was observed in the BIM error-generic setting, with only minimal numerical

differences. This is likely because, despite the random restarts introduced by PGD, in this scenario they do not provide a significant advantage, indicating that the adversarial examples found by a single initialization are already near-optimal. As a result, PGD effectively behaves like BIM in this setting, further highlighting the shared vulnerabilities of NN1's decision boundaries to iterative, gradient-based attacks.

Error-Specific Variant. Figure 6 shows the performance of NN1 under the error-specific PGD attack for different parameter values. In this case, increasing the perturbation budget ε caused a clear and progressive divergence between the overall accuracy and the targeted success rate. At the lowest tested value of $\varepsilon = 0.010$, the model remained robust, maintaining 86.2% accuracy and redirecting only 7.6% of samples to the target class. However, as ε grew, the targeted success rate rose rapidly: at $\varepsilon = 0.030$, nearly two-thirds of the samples (64.5%) were successfully redirected, and by $\varepsilon = 0.100$, the targeted success rate reached 83.5%, with only 9.9% overall accuracy remaining.

A similar pattern emerged when analyzing the effect of the step size ε_{step} . Starting from very low values, the model's accuracy was largely preserved (98.1% at $\varepsilon_{step} = 0.001$), and targeted success rate was negligible (1.0%). However, even modest increases in ε_{step} had dramatic effects: at $\varepsilon_{step} = 0.005$, the targeted success rate climbed to 16.8%, and at $\varepsilon_{step} = 0.010$ it soared to 64.4%, while accuracy simultaneously dropped to 23.8%. Beyond this point, additional increases in step size continued to amplify the targeted success rate, reaching 90.0% at $\varepsilon_{step} = 0.030$, with model accuracy bottoming out at 6.3%.

The number of iterations also proved to be an important factor. With just one iteration, the attack barely impacted the model (2.7% targeted success rate, 90.8% accuracy). Increasing to three iterations caused a sudden shift (64.0% targeted success rate, 24.0% accuracy), and by ten iterations, the attack fully converged (99.5% targeted success rate, 1.2% accuracy). Beyond this, addi-

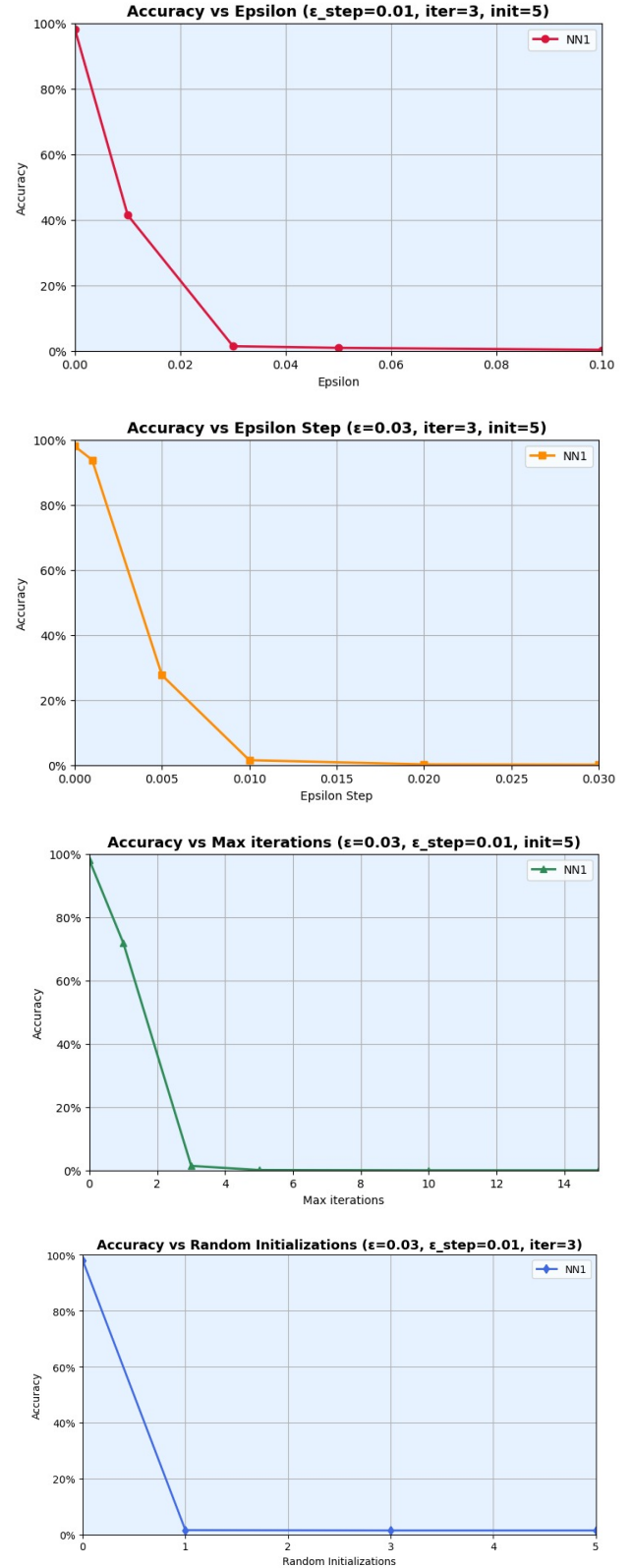


Figure 5: Security Evaluation Curves for PGD (error-generic variant) applied to NN1.

tional iterations had negligible effects, indicating rapid convergence once a sufficient number of steps was reached.

Finally, varying the number of random initializations showed a small but noticeable effect on both accuracy and targeted performance. While the overall accuracy and targeted success rate remained relatively stable across different values, there was a slight improvement as the number of restarts increased. For instance, accuracy decreased from 24.6% to 23.3% and the targeted success rate improved from 61.9% to 64.2% when moving from 1 to 5 restarts. This suggests that, in the error-specific setting, multiple random restarts can help to slightly refine the attack's effectiveness, enabling marginal improvements in targeted misclassification that would otherwise be missed.

This subtle yet measurable impact of random restarts likely explains why, unlike in the error-generic setting where random restarts had almost no effect and PGD's SECs closely matched those of BIM, here the error-specific performance profiles of PGD and BIM are not perfectly aligned. The small differences introduced by random restarts in PGD slightly shift the curves, highlighting how even minor variations can lead to meaningful divergence in more targeted attack scenarios.

Overall, these results highlight how the error-specific PGD attack becomes highly effective once a sufficient threshold is crossed in terms of ϵ , ϵ_{step} , or the number of iterations, with random restarts offering only a marginal refinement in performance.

3.3.5 DeepFool

DeepFool is an untargeted attack that iteratively perturbs an input sample by approximating the classifier's decision boundaries as linear hyperplanes. At each step, the algorithm identifies the nearest boundary and applies the minimal perturbation required to cross it, progressively refining the direction of the perturbation. The process continues until a misclassification is achieved or a maximum number of iterations is reached.

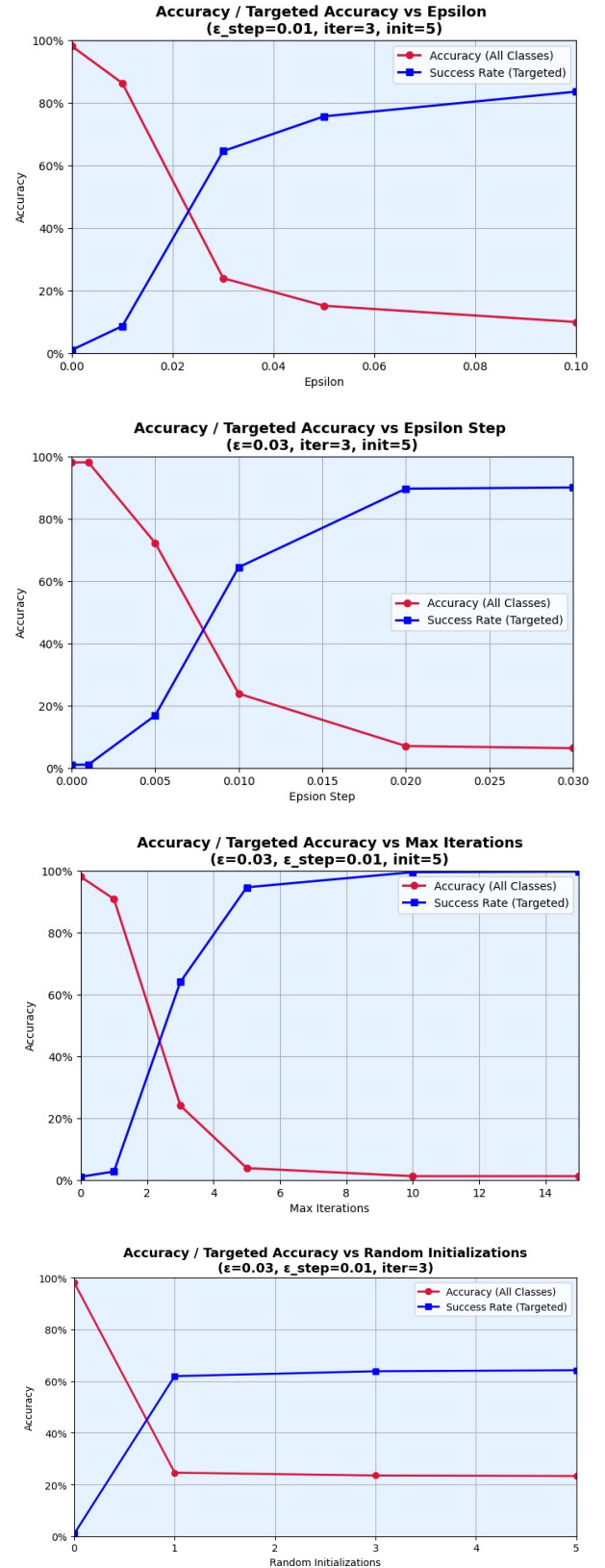


Figure 6: Security Evaluation Curves for PGD (error-specific variant) applied to NN1.

Two main parameters were varied to evaluate the attack's effectiveness: the number of iterations and the overshoot factor ε . Contrary to what one might expect, changes in the overshoot had a negligible impact on the final classification accuracy: regardless of whether ε was set to $1e-5$ or 0.005, the model's accuracy remained nearly constant (around 13.4%). This is because the overshoot is applied only after the iterative process concludes, serving as a small multiplicative margin to ensure that the perturbed sample crosses the decision boundary. Since the bulk of the perturbation is determined by the iterative steps, this parameter has limited influence unless set to extreme values.

In contrast, the number of iterations had a substantial effect on the attack's strength. With only one iteration, the model's accuracy dropped sharply from 98.6% to 13.4%, indicating that even minimal refinement was sufficient to generate impactful adversarial examples. Increasing the number of iterations further reduced accuracy, reaching 3.3% with just three iterations and as low as 0.6% with ten. However, despite its effectiveness, DeepFool consistently produced perturbations with ℓ_∞ norms exceeding 0.3 across all tested configurations, well above the maximum permitted value of 0.05. As such, while the attack demonstrates significant capacity to compromise the model, it fails to comply with the project's perturbation constraints and must therefore be excluded from the set of attacks considered valid within the scope of this project. The corresponding Security Evaluation Curves are shown in Figure 7.

3.3.6 | Carlini & Wagner L_∞

The Carlini & Wagner (C&W) attack is a powerful optimization-based method specifically designed to generate adversarial examples that are both highly effective and visually imperceptible. While its original formulation is typically implemented under the ℓ_2 norm, adaptations for the ℓ_∞ norm are also available and supported in **ART**. These variants optimize a surrogate loss to mislead the classifier, while simultaneously constraining the pertur-

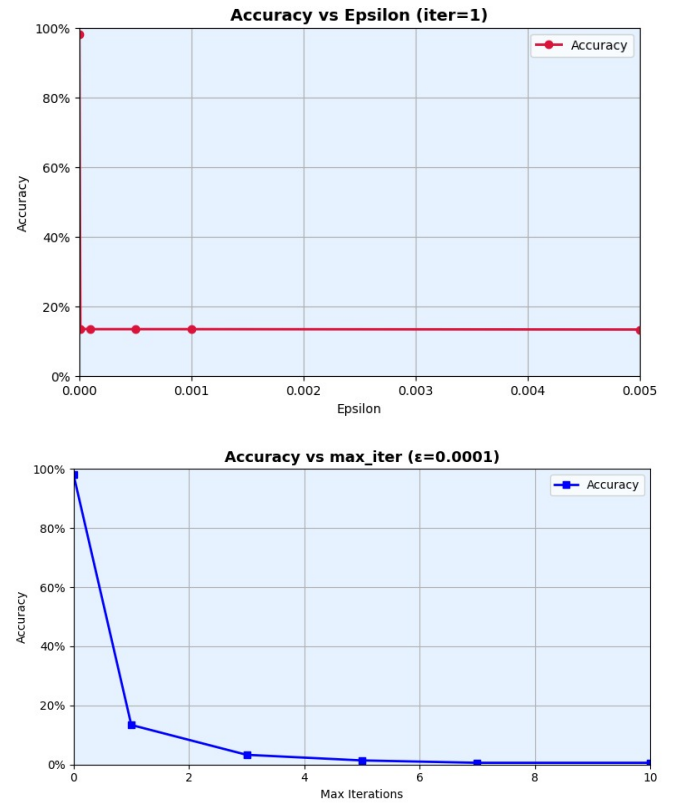


Figure 7: Security Evaluation Curves for DeepFool applied to NN1.

bation magnitude within a given ℓ_∞ budget.

Due to the computational intensity of this attack, especially under the ℓ_∞ setting, no full Security Evaluation Curves were produced. Even with optimizations in place, a single adversarial generation process required more than five hours of computation. As a result, only a limited number of configurations were tested for each variant.

Error-Generic Variant. When applied in the error-generic setting, the Carlini & Wagner L_∞ attack demonstrated substantial effectiveness, driving the model's top-1 accuracy down to just 0.2%. This outcome highlights the strength of this optimization-based method, even in the absence of extensive parameter tuning. The attack was carried out using a single viable configuration, selected for its relatively low computational cost: it employed a learning rate of 0.01, 7 optimization steps and an initial constant of $1e-5$ (capped at 1.0 and reduced

Metric	Value
Correctly classified	2
Misclassified	998
Accuracy (NN1)	0.20%
Max perturbation (ℓ_∞)	0.0200

Table 2: Classification performance of NN1 under the Carlini & Wagner L_∞ attack (error-generic variant).

by a factor of 0.5 during optimization). Although a full Security Evaluation Curve could not be generated, the observed results remain highly significant. Importantly, the perturbations respected the project's constraint on maximum distortion, remaining within the allowed ℓ_∞ norm of 0.05.

Error-Specific Variant. Due to the high computational cost of the Carlini & Wagner L_∞ attack, only one parameter configuration could be feasibly tested for the targeted setting. The chosen setup included a target confidence of 0.5, a learning rate of 0.01, 7 optimization steps and an initial constant of $1e-5$ (bounded above by 1.0 with a reduction factor of 0.5). This configuration was selected because it yielded the shortest execution time among all those considered.

Under these conditions, the model achieved a classification accuracy of 91.7% and a targeted success rate of 1%. Given that each identity in the test set accounts for exactly 1% of the total samples, this result indicates that no adversarial sample was effectively redirected to the target class. The small decrease in accuracy compared to the clean baseline (98.6%) suggests a limited destabilizing effect of the attack, without achieving its targeted objective.

The measured perturbation was also notably small, with a maximum ℓ_∞ norm of only 0.005, an order of magnitude below the allowed threshold of 0.05. This observation suggests that the limited effect of the attack may be due to insufficient perturbation strength. Attempts were made to explore more aggressive configurations to address this limitation; however, all such experiments were

Metric	Value
Correctly classified	917
Misclassified	83
Targeted success rate	1.00%
Accuracy (NN1)	91.70%
Max perturbation (ℓ_∞)	0.0050

Table 3: Classification performance of NN1 under the Carlini & Wagner L_∞ attack (error-specific variant).

rendered impractical due to excessive execution times.

4 | Transferability of Adversarial Examples

4.1 | Target Model: NN2 (ResNet50)

To assess the transferability of adversarial examples, a second model was employed alongside NN1. This model, referred to as NN2, is based on the **ResNet50 architecture** and corresponds to the version named `resnet50_scratch`, which was provided pre-trained on the **VGGFace2 dataset** as part of the implementation available in the open-source repository by cydonia999 [1]. The model was used as-is, without modification or re-training. Unlike NN1, which outputs 512-dimensional embeddings and requires a separate classification layer, NN2 directly performs classification over the full set of identities it was trained on. In particular, the network includes a final fully connected layer that maps the extracted features to class logits, allowing it to operate as a closed-set face classifier.

Input images were processed using the same **MTCNN** pipeline adopted for NN1, which detects and aligns the face region and directly returns RGB crops at the required resolution of 224×224 pixels. Before being passed to the network, these images were converted to BGR format and normalized by subtracting a fixed per-channel mean (91.4953 for blue, 103.8827 for green, and

131.0912 for red), in accordance with the preprocessing procedure defined in the original implementation of the model.

Although both NN1 and NN2 were trained on the same dataset, the two models differ significantly in their internal structure and in how facial features are represented and processed. These differences make NN2 a suitable candidate for evaluating the robustness and transferability of adversarial perturbations across architectures.

4.2 | Baseline Accuracy on Clean Data

To establish a performance baseline for NN2, we evaluated the model on the same clean test set used for NN1, consisting of 1000 images equally distributed among 100 distinct identities. As in the case of NN1, top-1 accuracy was used to assess how often the model's highest-scoring prediction matched the correct identity.

NN2 correctly classified 987 out of 1000 samples, resulting in an overall accuracy of 98.70%. This performance is slightly higher than that of NN1 (98.60%), despite the architectural differences between the two models. Of the 13 misclassified samples, most errors occurred only once and appeared to be randomly distributed across identities.

One notable exception was the identity *Magdalena Gessler*, which was misclassified twice, once as *Rocío Jurado* and once as *Judit Mascó*. This mirrors the behavior observed with NN1, where the same identity also accounted for multiple errors, possibly indicating particular visual features that make her harder to distinguish.

Overall, NN2 demonstrated strong and consistent performance under clean conditions, providing a reliable foundation for subsequent transferability analysis.

Metric	Value
Total test samples	1000
Correctly classified	987
Misclassified	13
Accuracy on clean data (NN2)	98.70%

Table 4: Classification performance of NN2 on the clean test set.

4.3 | Transfer Evaluation Protocol

To evaluate the transferability of adversarial attacks, the examples originally crafted against NN1 were subsequently tested on NN2. This allowed us to investigate whether perturbations effective on one model could generalize to a different architecture trained on the same dataset.

Due to architectural and preprocessing differences between NN1 and NN2, a direct reuse of the adversarial samples was not immediately feasible. Adversarial examples generated for NN1 are 160×160 RGB images with pixel values normalized in the range $[-1, 1]$, consistent with the input requirements of the **Inception ResNet V1** architecture. In contrast, NN2 expects 224×224 images formatted in BGR order and normalized according to a fixed per-channel mean subtraction.

To enable a fair comparison, the adversarial images were first resized to match NN2's expected resolution and then converted to BGR format. Finally, the appropriate normalization was applied prior to inference. These preprocessing steps ensured compatibility without altering the adversarial perturbations, thereby preserving the integrity of the transferability analysis.

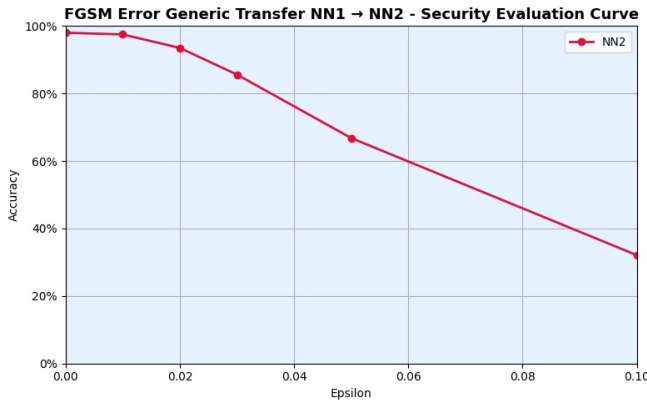


Figure 8: Security Evaluation Curve for FGSM (error-generic variant) applied to NN2.

4.4 | Attacks Results

4.4.1 | Fast Gradient Sign Method (FGSM)

Error-Generic Variant. Error-generic FGSM perturbations originally crafted to fool NN1 were evaluated on NN2 to assess their impact under transfer. As illustrated in Figure 8, accuracy dropped progressively from 98.7% on clean data to 97.5% at $\epsilon = 0.010$, and continued to decline as the perturbation strength increased, reaching 32.1% at $\epsilon = 0.100$. Although the attack was not tailored to NN2, the degradation in performance reveals a clear degree of transferability.

This behavior stands in contrast to NN1, where the same adversarial examples caused a much sharper decline. For instance, at $\epsilon = 0.050$, NN1's accuracy fell to just 5%, while NN2 still retained 66.8%. At the maximum tested perturbation of $\epsilon = 0.100$, NN1 had nearly collapsed (1.5% accuracy), whereas NN2 remained substantially more resilient. These results suggest that FGSM perturbations, though transferable, lose some of their effectiveness when applied across architectures.

Error-Specific Variant. When transferred to NN2, targeted FGSM attacks exhibited only limited effectiveness. As shown in Figure 9, the success rate in redirecting pre-

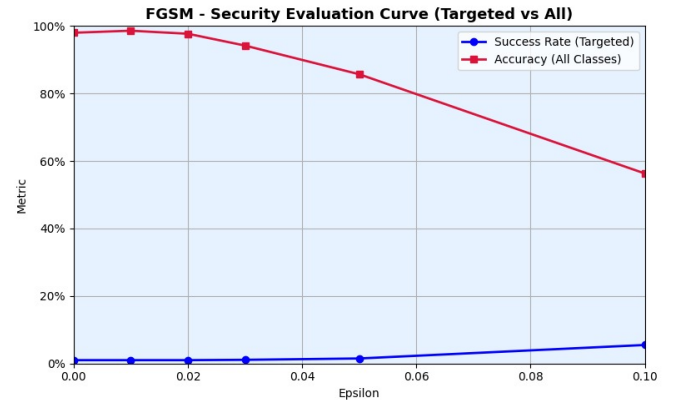


Figure 9: Security Evaluation Curve for FGSM (error-specific variant) applied to NN2.

dictions toward the target class remained consistently low, increasing only marginally from 1.0% at $\epsilon = 0.010$ to 5.5% at $\epsilon = 0.100$. This is in stark contrast to the behavior observed on NN1, where the success rate rose steadily and reached 47.1% for the same maximum perturbation.

Despite the weak targeted effect, the general accuracy of NN2 still decreased with increasing ϵ , dropping from 98.7% to 56.3%. This suggests that while the perturbations disrupted the classifier's decisions, they failed to consistently bias the output toward the chosen class. Compared to NN1, where both the overall accuracy and targeted success rate showed a clear and correlated decline, NN2 appears far more resistant to the specific directional influence of FGSM perturbations.

4.4.2 | Basic Iterative Method (BIM)

Error-Generic Variant. To assess the transferability of the BIM untargeted attack, adversarial examples generated on NN1 were evaluated against NN2. As illustrated in Figure 10, NN2 showed substantially greater robustness. While NN1's accuracy plummeted to just 0.5% at $\epsilon = 0.030$, NN2 retained a high accuracy of 79.0% under the same conditions, and no further drop was observed for larger ϵ values, indicating an early saturation in the

impact of the attack on NN2.

A similar pattern emerged when varying the step size ϵ_{step} : for NN1, increasing the step size from 0.001 to 0.010 resulted in a dramatic drop from 94.2% to 0.5%. In contrast, NN2's accuracy decreased much more gradually, from 98.7% to 79.0%, and remained above 76.0% even with larger steps, highlighting a significantly higher tolerance to per-step perturbations.

Varying the number of iterations produced only minor fluctuations in NN2's performance. Unlike NN1, where accuracy dropped to 0% after a few iterations, NN2 exhibited a more complex pattern: while accuracy initially fell sharply (from 97.5% to 74.5%), it then showed a slight recovery at higher iteration counts (rising to 79.2% at fifteen iterations). This suggests that additional attack steps beyond a certain point may no longer contribute to destabilizing the model effectively and can even cause adversarial examples to re-enter regions of the decision space where the classifier regains some confidence.

Overall, this confirms NN2's higher resilience and the fact that adversarial perturbations optimized on NN1 do not always translate to sustained misclassification on NN2, especially for higher iteration counts.

Error-Specific Variant. The targeted version of BIM exhibited significantly lower effectiveness when transferred to NN2. As shown in Figure 11, the targeted success rate remained constant at just 1% across all values of ϵ , ϵ_{step} , and number of iterations. This flat trend indicates a complete failure to redirect inputs toward the chosen target class, regardless of the attack strength.

Unlike what was observed on NN1, where increasing any of the main parameters led to a sharp rise in targeted success rate and a collapse in general accuracy, NN2 remained largely unaffected in both respects. The top-1 accuracy consistently stayed above 93%, showing only minor fluctuations as parameters varied. Notably, this robustness persisted even when parameters that had proven decisive on NN1 were applied, confirming the inability of transferred adversarial examples to enforce targeted misclassification on NN2.

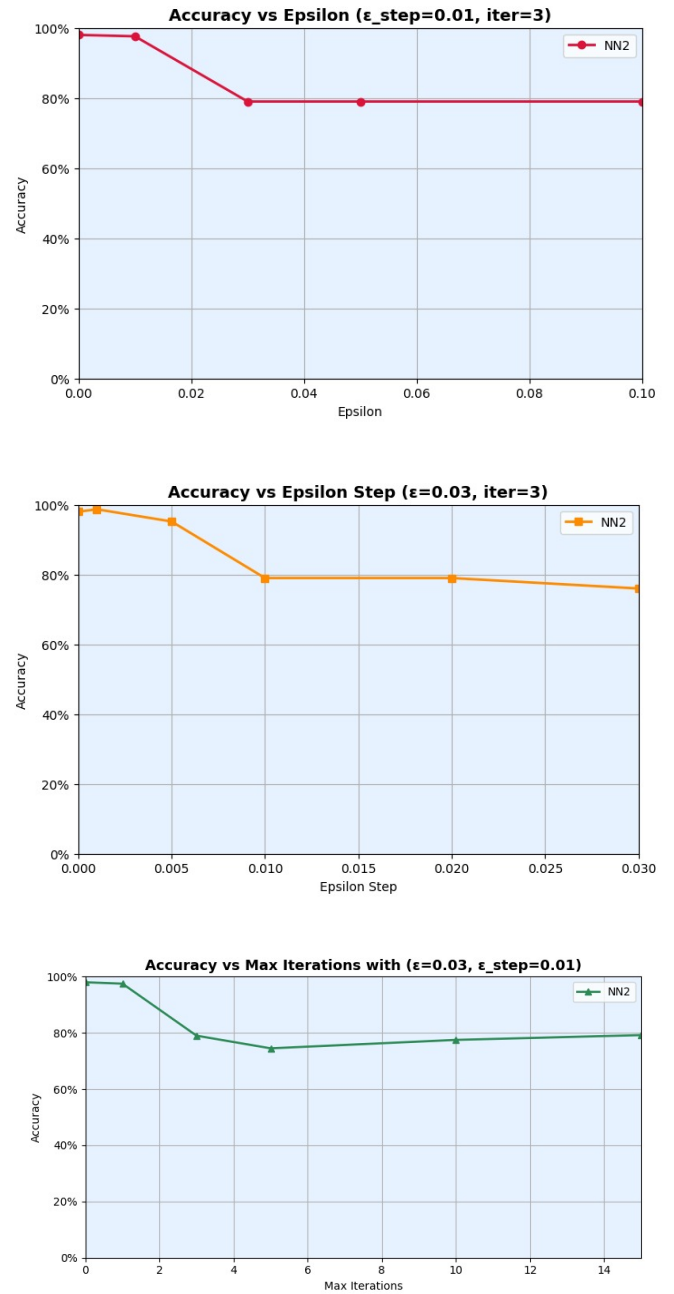


Figure 10: Security Evaluation Curves for BIM (error-generic variant) applied to NN2.

These results highlight a stark contrast between the two models: while NN1 was highly vulnerable to targeted BIM attacks, NN2 showed strong resistance, maintaining stable predictions and effectively nullifying the intended redirection effect of the adversarial inputs.

4.4.3 Projected Gradient Descent (PGD)

Error-Generic Variant. Figure 12 shows the performance of NN2 under error-generic PGD attacks generated from NN1. As the perturbation budget ϵ increases, the model's accuracy declines gradually from 97.6% at $\epsilon = 0.010$ to 80.6% at $\epsilon = 0.100$. This degradation is smooth and contained, with the model maintaining strong classification performance even at the highest perturbation level. When compared to NN1, the difference is substantial: at $\epsilon = 0.030$, NN1's accuracy had already collapsed to 1.4%, and dropped below 1% for higher values. This stark contrast highlights NN2's significantly greater robustness to PGD attacks transferred from NN1.

A similar behavior is observed when varying the step size ϵ_{step} . NN2's accuracy drops from 96.3% to 78.5% as the step increases from 0.005 to 0.030, but this decline is far milder than the one observed for NN1, which reached near-zero accuracy over the same range.

Regarding the number of iterations, NN2's accuracy declines from 88.7% at three iterations to 78.0% at ten, with a slight rebound to 79.1% at fifteen iterations. This suggests that beyond a certain point, additional steps no longer increase the attack's effectiveness and may even allow the classifier to partially recover its predictions, similar to what was observed with the error-generic variant of BIM.

Finally, varying the number of random initializations led to only minor fluctuations in NN2's performance, never exceeding around 1%. This suggests that the overall stability of NN2 is maintained regardless of the initial random conditions, highlighting its robustness against

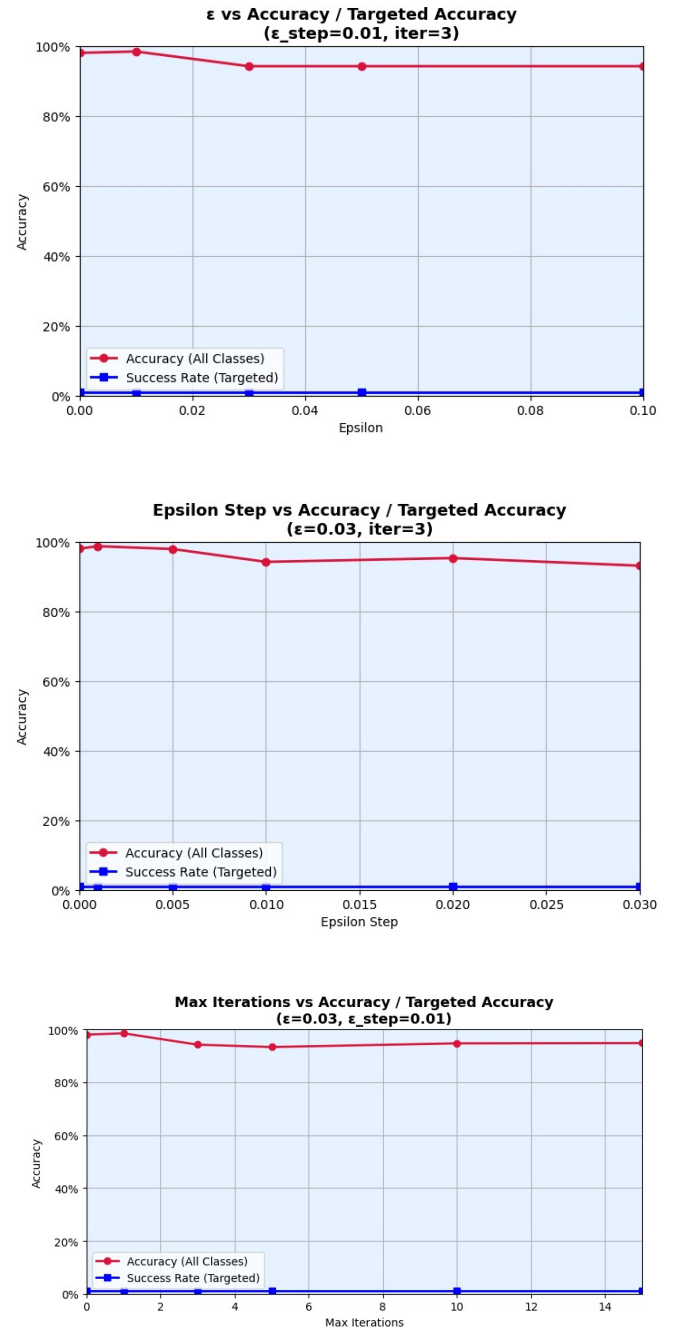


Figure 11: Security Evaluation Curves for BIM (error-specific variant) applied to NN2.

these subtle attack variations.

In conclusion, while PGD remains a strong attack strategy, NN2 exhibits a markedly higher resilience to these perturbations. Its ability to retain high classification accuracy across a range of attack strengths and configurations confirms its superior robustness compared to NN1 in the error-generic transfer scenario.

Error-Specific Variant. The targeted variant of PGD yielded drastically different results when evaluated on NN2 compared to NN1. Despite varying the main parameters, ϵ , ϵ_{step} , max iterations, and the number of random initializations, the attack consistently failed to redirect predictions toward the intended target class. As shown in Figure 13, the targeted success rate remained fixed at just 1% across all tested configurations, including highly permissive ones such as $\epsilon = 0.100$ or fifteen iterations.

While accuracy did experience a mild decline in response to more aggressive settings, dropping from 98.6% to 94.8% as ϵ increased, and similarly decreasing with higher step sizes, iteration count and number of random initializations, the general degradation was incredibly limited, especially when compared to NN1. For instance, with $\epsilon = 0.030$, NN1 showed a targeted success rate above 60% and accuracy below 25%, whereas NN2 preserved over 96% accuracy with no meaningful improvement in redirection success.

These results highlight a significant robustness advantage of NN2 in the targeted setting: unlike NN1, which proved highly vulnerable once a critical parameter threshold was reached, NN2 maintained both classification integrity and resistance to class redirection, even under extensive PGD configurations. This suggests that, at least for this model and task, targeted adversarial perturbations generated on NN1 fail to generalize in a way that meaningfully compromises NN2's predictions.

4.4.4 | Carlini & Wagner L_∞

Error-Generic Variant. As part of the transfer evaluation, NN2 was tested against the error-generic

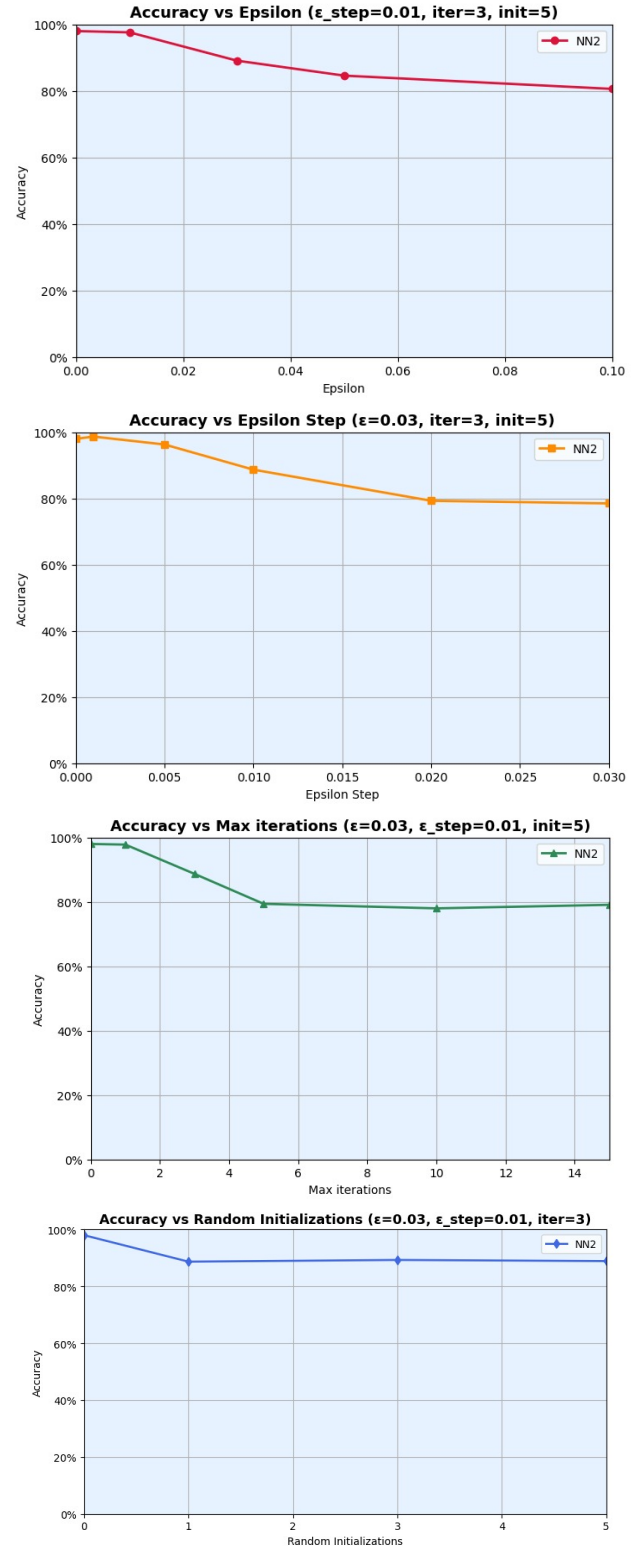


Figure 12: Security Evaluation Curves for PGD (error-generic variant) applied to NN2.

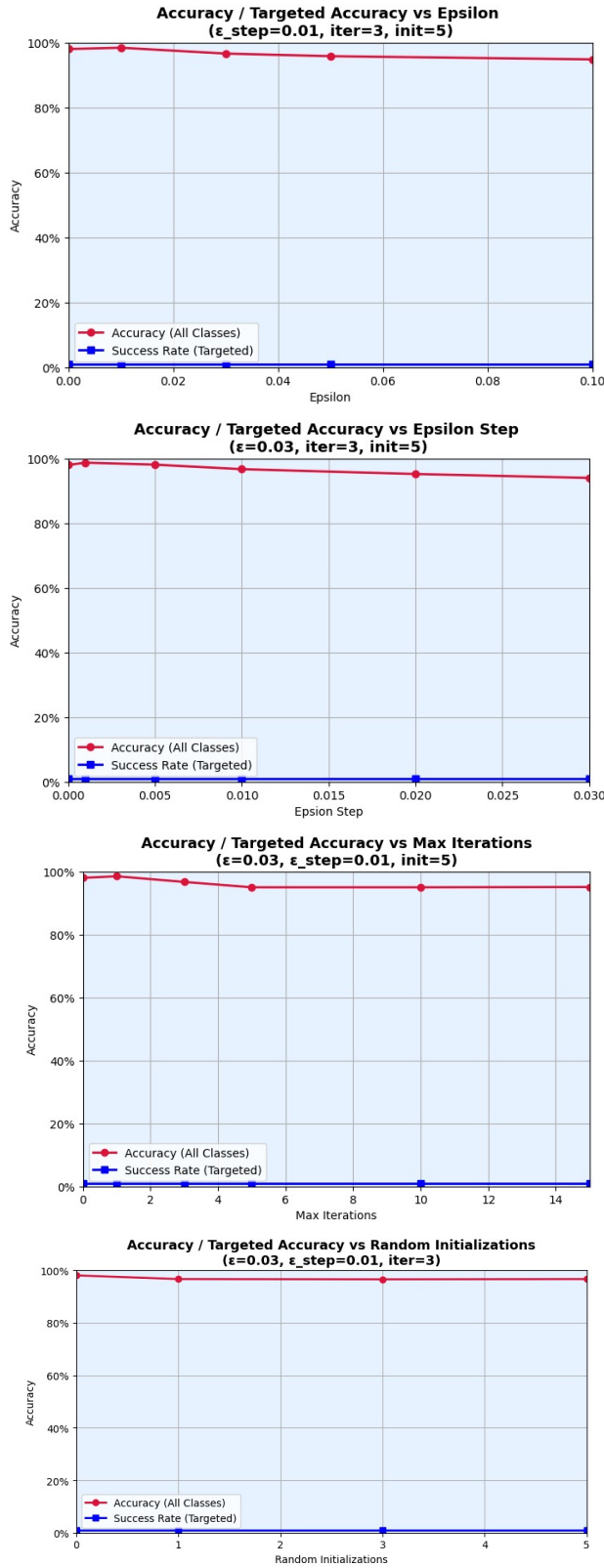


Figure 13: Security Evaluation Curves for PGD (error-specific variant) applied to NN2.

Metric	Value
Correctly classified	973
Misclassified	27
Accuracy (NN2)	97.30%
Max perturbation (ℓ_{∞})	0.020

Table 5: Classification performance of NN2 under the Carlini & Wagner L_{∞} attack (error-generic variant).

Metric	Value
Correctly classified	983
Misclassified	17
Targeted success rate	1.00%
Accuracy (NN2)	98.30%
Max perturbation (ℓ_{∞})	0.0050

Table 6: Classification performance of NN2 under the Carlini & Wagner L_{∞} attack (error-specific variant).

Carlini & Wagner L_{∞} adversarial examples previously crafted for NN1 under a single parameter setting. As previously discussed, this constraint was imposed by the considerable computational burden of the attack, which made broader experimentation impractical.

While this configuration proved highly effective against NN1, reducing its top-1 accuracy to just 0.2%, the impact on NN2 was markedly less pronounced.

Specifically, NN2 preserved a top-1 accuracy of 97.3%, only marginally below its clean baseline of 98.7%. This minimal degradation suggests that the perturbations had very limited disruptive power when transferred to this model. The attack failed to induce any significant misclassification behavior, highlighting the stronger robustness of NN2 in this setting.

Error-Specific Variant. As previously mentioned, due to the high computational demands of the error-specific Carlini & Wagner L_{∞} attack, only a single parameter configuration was evaluated on NN1. Consequently, the transferability analysis to NN2 was conducted using only the adversarial examples generated under that same setup.

When tested on NN2, the results mirrored those observed on NN1. The model maintained a high classification accuracy of 98.3%, with a targeted success rate of only 1.0%. Since each identity constitutes 1% of the dataset, this indicates that no sample was successfully redirected to the designated target class.

Compared to NN1, NN2 appears equally resilient to this attack, if not slightly more so, given the even smaller drop from its clean accuracy baseline of 98.7%. As previously discussed, the weak performance is likely attributable to the limited perturbation magnitude (0.0050), but more aggressive configurations could not be tested due to impractical execution times.

4.5 | Conclusions on Transferability Analysis

Overall, the transferability analysis shows that NN2 is considerably more robust to adversarial examples crafted on NN1. In the error-generic setting, while NN2 did experience some accuracy degradation, the drop was much less severe compared to the dramatic collapse observed in NN1. This trend is even more pronounced in the error-specific setting, where the targeted success rate remained consistently around 1%, and the impact on accuracy was minimal. These observations demonstrate that architectural and preprocessing differences in NN2 offer substantial resilience to these perturbations generated to fool NN1.

5 | Defensive Strategies against Adversarial Attacks

In this final part of the project, we focused on implementing and evaluating defense strategies against adversarial attacks. Since NN1 exhibited a significant vulnerability to these attacks, we needed to explore ways to further strengthen the system's resilience. Our efforts concentrated on two main directions: a preprocessing-based de-

fense, involving techniques like JPEG compression, local smoothing, and colour bits squeezing, and an adversarial detector trained to catch gradient-based attacks.

For the preprocessing defense, we conducted experiments on both clean and adversarial data, tuning the parameters to achieve a balance between robustness and preservation of clean-data performance. Although the results showed some improvement, especially for more sophisticated attacks like Carlini & Wagner, this approach was less effective for simpler attacks such as FGSM. Therefore, we developed an adversarial detector to complement the preprocessing, aiming for a combined defense mechanism.

This section outlines the design and evaluation of both defenses, illustrating their impact on the model's performance under various attack scenarios and explaining how they can be integrated for better protection in practice.

5.1 | Preprocessing-based Defense

5.1.1 | Defense Approach and Implementation

For the preprocessing-based defense of our NN1 model, we adopted the concept of feature squeezing, a strategy introduced by Xu et al.[3]. This approach is grounded in the idea that the input feature space for images is often much larger than necessary, providing adversarial attacks with excessive room to manipulate inputs. By intentionally reducing the complexity of the input space, we can constrain these opportunities and improve the model's resilience.

In line with the methods described in Xu et al.[3], we implemented bit-depth reduction and median filtering through the **ART** library. Bit-depth reduction involves decreasing the color depth of images, essentially reducing the number of bits used to represent each pixel. This process simplifies the representation of input data and

makes it more difficult for small perturbations to remain effective. Median filtering, a local smoothing technique, replaces each pixel's value with the median of its surrounding neighborhood. This method helps eliminate minor variations and adversarial noise by enforcing local consistency.

To complement these methods, we incorporated JPEG compression, inspired by the work of Dziugaite et al.[2]. JPEG compression can be seen as another form of feature squeezing, as it removes high-frequency components from the image. Through compression and decompression, this technique diminishes subtle adversarial perturbations that rely on slight, high-frequency changes to the input data.

We decided to combine all three of these preprocessing techniques after evaluating their individual impact on adversarial attacks and considering the insights provided by Xu et al.[3]. Although our dataset, **VGGFace2**, was not directly tested in their study, the presence of color images made it applicable to the same principles. Our experiments involved testing these defenses against gradient-based attacks (error-generic and error-specific) and the Carlini & Wagner attack. For the latter, as noted in previous sections, we only explored a single parameter setting due to computational limitations.

This combined approach aims to reduce the effectiveness of adversarial attacks while preserving the model's accuracy on clean data. The next sections will describe how we selected the parameters for each technique and will present the results of our experiments on clean and adversarial samples.

5.1.2 Experimental Analysis

To evaluate the effectiveness of our preprocessing defense for NN1, we conducted a thorough experimental analysis covering parameter selection, performance on clean data, and performance on adversarial data. The goal was to find parameter settings that would improve adversarial robustness while preserving the model's accuracy on clean samples. While this section primarily

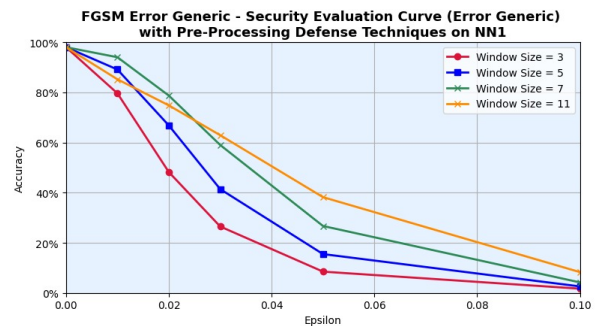


Figure 14: Security Evaluation Curves for FGSM (error-generic variant) with Local Smoothing.

presents plots for the error-generic scenarios, we also conducted the same tests for the error-specific cases. In all these instances, the observed trends and outcomes were similar, so we chose to show only the generic results as they effectively represent the overall behavior.

Initially, we analyzed the impact of each preprocessing technique by applying it separately to adversarial examples generated with different perturbation magnitudes. Although we chose to include only the plots for FGSM attacks in this report for brevity, we carried out the same tests for BIM and PGD as well. These attacks exhibited trends very similar to FGSM: the accuracy curves shifted in the same way, indicating that the findings generalize across all three attacks.

For spatial smoothing, we experimented with window sizes ranging from 3 to 11, observing that increasing the window size enhances the defense capability by reducing small variations in the image, but also introduces a trade-off with respect to clean-data performance. In particular, window size 7 emerged as a reasonable balance between smoothing out adversarial noise and preserving important features of the original images (Figure 14).

For bit-depth reduction, we tested values of 1, 4, and 5 (Figure 15). As anticipated from the results reported by Xu et al. [3] for color image datasets (like **VGGFace2**), reducing the bit depth to 1 caused a severe degradation in performance on clean images. Between bit depths 4 and 5, the differences in adversarial robustness were minimal, but bit depth 5 preserved more color information

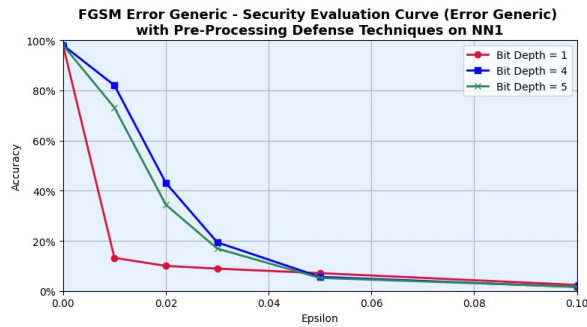


Figure 15: Security Evaluation Curves for FGSM (error-generic variant) with Bit Depth Reduction.

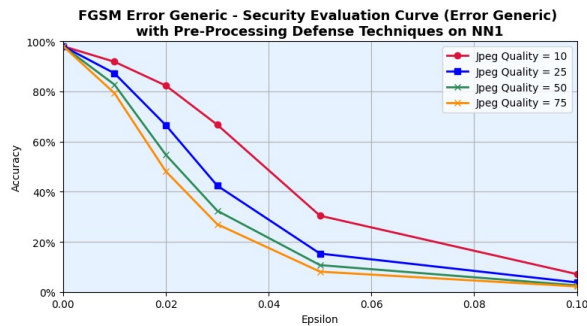


Figure 16: Security Evaluation Curves for FGSM (error-generic variant) with JPEG Compression.

in the input images, making it our preferred choice.

JPEG compression was evaluated with quality factors 10, 25, 50, and 75 (Figure 16). While a lower quality factor like 10 can remove adversarial noise more effectively, it also substantially degrades the clean image. A quality factor of 25 provided the best compromise, maintaining visual fidelity while still removing high-frequency perturbations.

After analyzing the individual impact of each preprocessing technique, we visualized and compared two combined parameter configurations in our final evaluation. The first configuration was chosen as the best trade-off between adversarial robustness and preservation of clean-data accuracy (window size 7, bit depth 5, JPEG quality 25), resulting in a small drop from 98.6% to 98.3% in clean-data accuracy (Table 7). The second configuration was instead determined by focusing purely on adversarial robustness, choosing parameters that, when

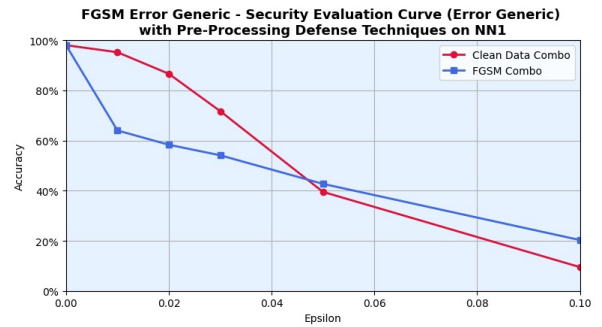


Figure 17: Security Evaluation Curves for FGSM (error-generic variant) with Full Preprocessing Pipeline.

tested separately, showed the best results on adversarial data alone (window size 11, bit depth 4, JPEG quality 10).

For FGSM attacks, as shown in Figure 17, the two combinations produce noticeably different behaviors. The red curve, corresponding to the clean-data optimized configuration, maintains higher accuracy for smaller perturbation magnitudes but declines more rapidly as epsilon increases. In contrast, the blue curve, based on adversarial-data optimization, offers slightly better performance at larger perturbations but sacrifices more clean-data accuracy overall. These two curves were included to clearly illustrate how different parameter choices can lead to different accuracy trends under FGSM attacks, even though the final defense configuration for further testing remained the one balancing both clean and adversarial performance.

For BIM and PGD attacks, the results showed an unexpected trend: the clean-data optimized configuration consistently outperformed the adversarial-optimized configuration across all tested perturbation magnitudes. As shown in Figure 18, for BIM the red curve, representing the clean-data optimized combination, remains above the blue one, even for larger perturbation levels. Although we initially expected the adversarial-optimized configuration to perform better for these stronger attacks, it did not provide any advantage in our experiments. This outcome highlights how the performance observed when optimizing for adversarial data alone do not

Technique	Parameter	Clean Accuracy (%)
JPEG Compression	q=10	97.2
	q=25	98.4
	q=50	98.5
	q=75	98.4
Feature Squeezing	1-bit	13.7
	4-bit	98.3
	5-bit	98.5
Spatial Smoothing	w=3	98.6
	w=5	98.7
	w=7	98.5
	w=11	90.6
Best Overall Combination		98.7
Attacks-Optimized Combination		67.5

Table 7: NN1 clean-data accuracy (%) for different preprocessing techniques and parameter values.

Technique	Parameter	Accuracy (%)
JPEG Compression	q=25	38.7
Feature Squeezing	5-bit	5.3
Spatial Smoothing	w=7	93.4
Combined Defense	(w=7, 5-bit, q=25)	94.1

Table 8: NN1 accuracy (%) on Carlini & Wagner error-generic for different preprocessing techniques.

always translate into improved robustness when techniques are combined.

For the Carlini & Wagner attack, which had reduced the NN1’s accuracy to 0.2% without any defense, the same clean-data optimized preprocessing pipeline led to a significant improvement, raising accuracy to 94.1%. While spatial smoothing alone contributed the most (93.4%), the addition of bit-depth reduction and JPEG compression further reduced the number of misclassifications by seven. In the error-specific scenario of this attack, the defense also increased accuracy from 91.7% to 97.7%, demonstrating the particular effectiveness of this combined approach for this type of perturbation.

Technique	Parameter	Accuracy (%)
JPEG Compression	q=25	94.4
Feature Squeezing	5-bit	92.3
Spatial Smoothing	w=7	97.7
Combined Defense	(w=7, 5-bit, q=25)	97.9

Table 9: NN1 accuracy (%) on Carlini & Wagner error-specific for different preprocessing techniques.

5.2 Adversarial Detector

While the preprocessing-based defense significantly improved the model’s robustness against the Carlini & Wagner attack and iterative gradient-based attacks such as BIM and PGD, we noticed that its effectiveness for simpler attacks like FGSM was more limited, especially as perturbation magnitude increased. Furthermore, for BIM and PGD there was still a substantial drop in accuracy compared to the clean baseline, suggesting room for improvement. For these reasons, we decided to complement the preprocessing strategy with a second line of defense: a binary adversarial detector.

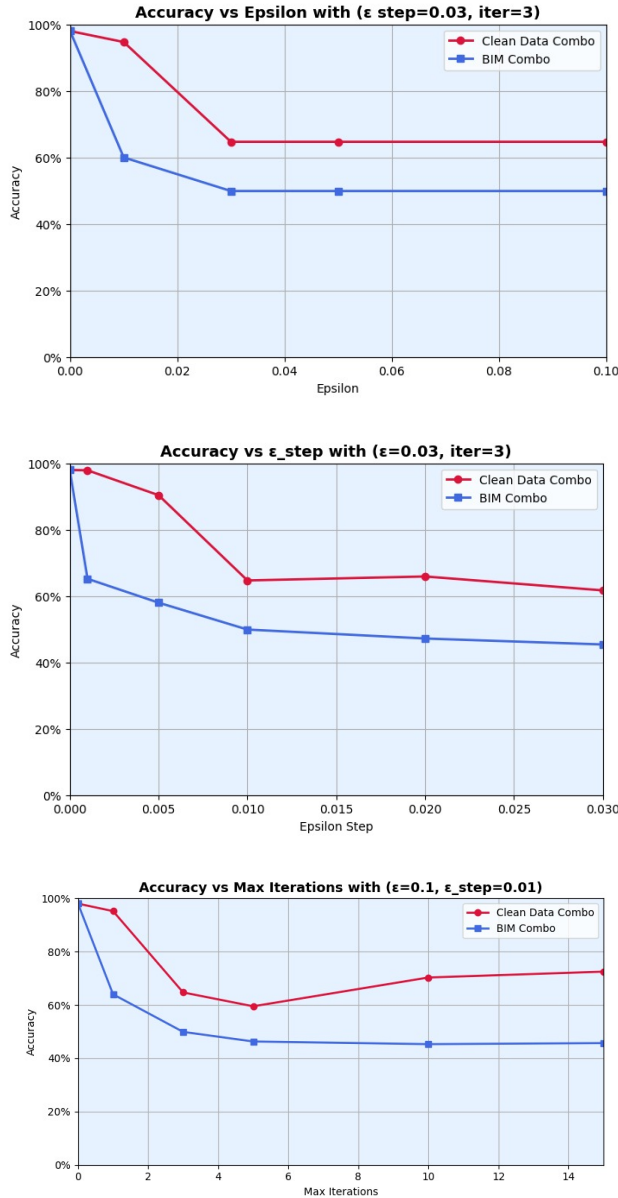


Figure 18: Security Evaluation Curves for BIM (error-generic variant) with Full Preprocessing Pipeline.

This detector was implemented using a **ResNet50 architecture**, whose final linear layer was replaced by a two-class output layer for binary classification between clean and adversarial examples. We selected **ResNet50** after preliminary tests showed it achieved the best performance among the architectures evaluated, given the same training parameters and dataset.

To train the detector, we constructed a balanced dataset of 2000 clean samples and 2000 adversarial samples. The adversarial examples included 750 generated with FGSM, 750 with PGD, and 500 with BIM, created using different parameter values within the project's constraints. We fine-tuned not only the final linear layer of the network but also the preceding layer, which proved to be important for adapting the model to this new binary classification task.

The detector reached an accuracy of 99.6% on the clean test set of 1000 samples, with only four clean images misclassified as adversarial, confirming its reliability on benign data. We also evaluated its performance against each specific attack type to provide a comprehensive picture of its behavior under different adversarial scenarios. In the following paragraphs, we present and comment on the detection accuracy curves for FGSM, BIM, and PGD attacks, highlighting how the detector responds to varying perturbation strengths.

FGSM Attacks For FGSM attacks, both error-generic and error-specific, the security evaluation curves (Figures 20) show that starting from a perturbation magnitude of $\epsilon = 0.02$, the detector is able to identify nearly all adversarial samples. Even for lower perturbation magnitudes, the number of adversarial samples detected consistently exceeds the number of samples misclassified by the model. This suggests that, although not all adversarial samples are detected below $\epsilon = 0.02$, the detector still identifies enough of them to support the classifier's decision-making process. In practice, this means that the detector can effectively help mitigate the impact of FGSM adversarial examples, especially as the perturbation strength increases.

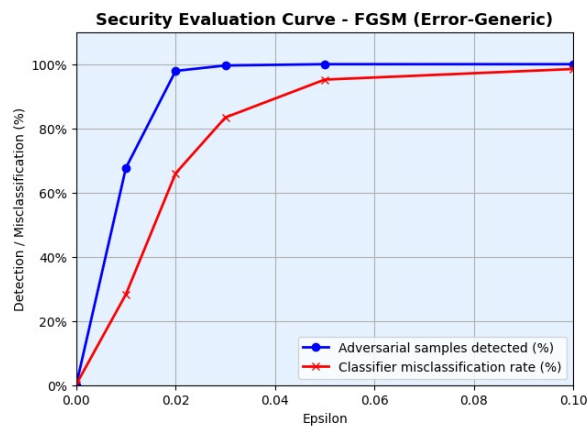


Figure 19: Security Evaluation Curves for FGSM (error-generic variant) with Detector Accuracy.

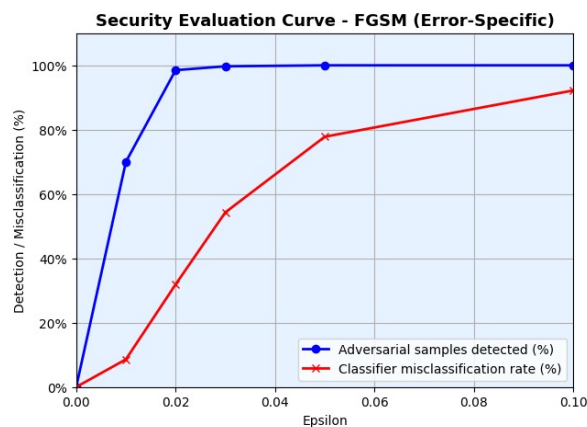


Figure 20: Security Evaluation Curves for FGSM (error-specific variant) with Detector Accuracy.

BIM Attacks For BIM attacks, the security evaluation curves reveal a striking overlap between the detection rate and the classifier's misclassification rate in the error-generic scenario (Figure 21). The two curves closely follow each other as the attack strength increases, whether by raising the number of iterations, the epsilon step, or the epsilon magnitude, and they both approach 100% very quickly. This nearly perfect overlap suggests that whenever the classifier begins to fail on adversarial samples, the detector consistently flags those inputs as adversarial.

In the error-specific scenario (Figure 22), the two curves remain similar but do not coincide as perfectly.

Here, the blue curve (detector) consistently remains above the red curve (misclassification) at lower perturbation levels, meaning that the detector is able to flag more adversarial samples than the classifier actually misclassifies. This behavior shows that even when the perturbation is subtle and the classifier's accuracy is still relatively high, the detector is already catching a significant portion of adversarial inputs. As the perturbation magnitude grows, both curves reach 100%, showing that the detector fully identifies all adversarial examples in these stronger attacks.

PGD Attacks For PGD attacks, the detector demonstrates solid performance in both error-generic and error-specific scenarios. In the error-generic case (Figure 23), the detection rate (blue curve) and the classifier's misclassification rate (red curve) rise together very quickly. After minimal increases in random initializations, epsilon step size, or iteration count, both curves reach 100% and stay perfectly aligned. This alignment suggests that once adversarial perturbations become strong enough to affect the classifier's decisions, the detector also fully identifies those inputs as adversarial. Even at the smallest perturbation levels, the detector generally tracks closely with the classifier's errors, leaving little room for adversarial examples to bypass detection in this setting.

In contrast, the error-specific scenario (Figure 24) shows a different pattern. Here, the detector's detection rate (blue curve) consistently exceeds the classifier's misclassification rate (red curve), especially at lower perturbation values. This means that the detector can identify more adversarial samples than those that actually lead to misclassifications, which is particularly important when the classifier still performs well and adversarial perturbations remain subtle. As the attack intensity increases, through more iterations, higher epsilon, or more random initializations, both curves converge towards 100%.

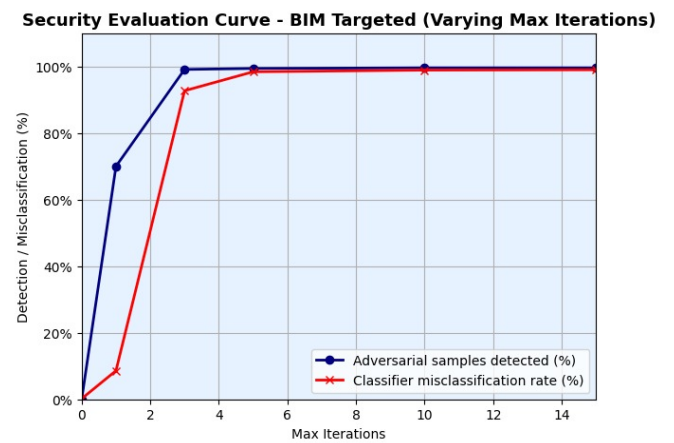
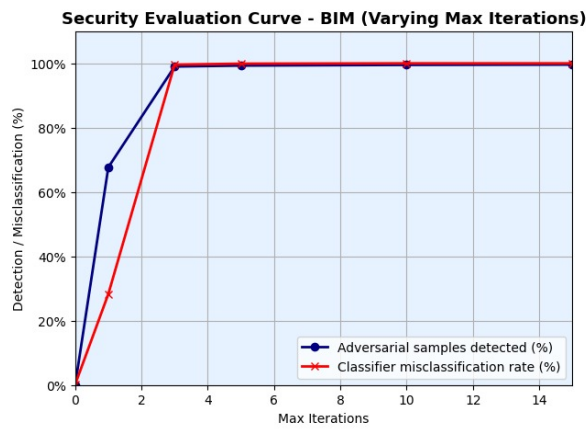
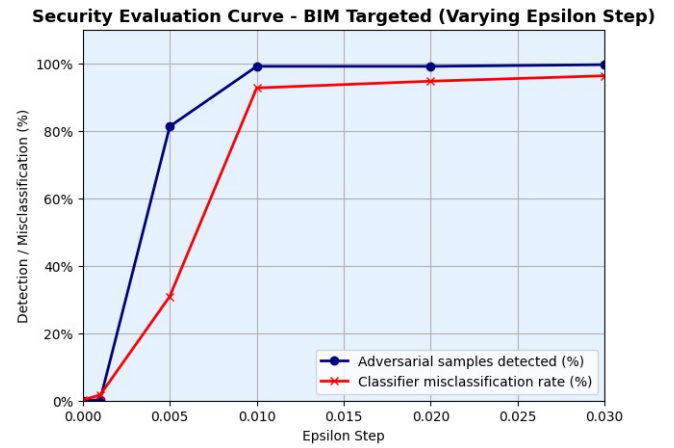
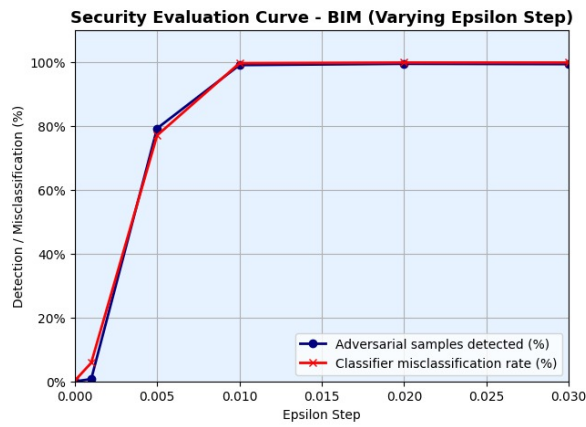
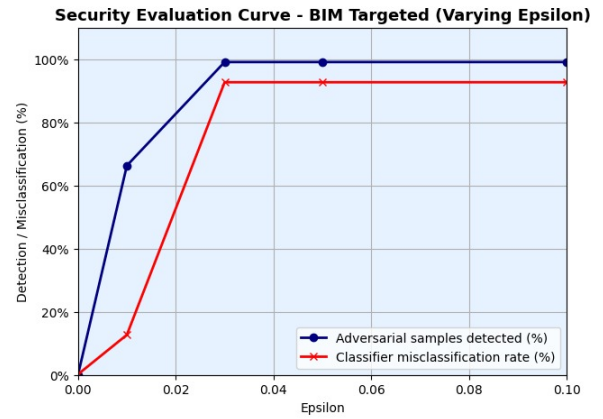
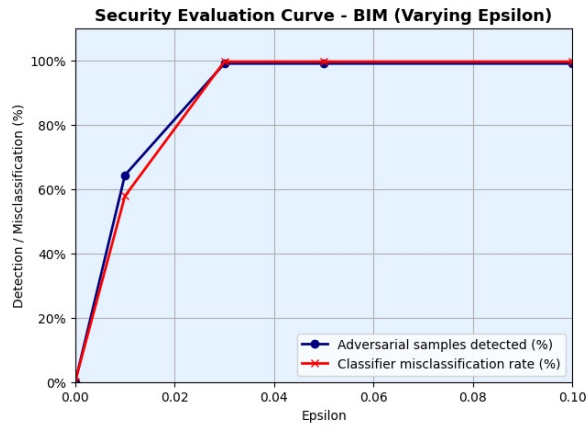


Figure 21: Security Evaluation Curves for BIM (error-generic variant) with Detector Accuracy.

Figure 22: Security Evaluation Curves for BIM (error-specific variant) with Detector Accuracy.

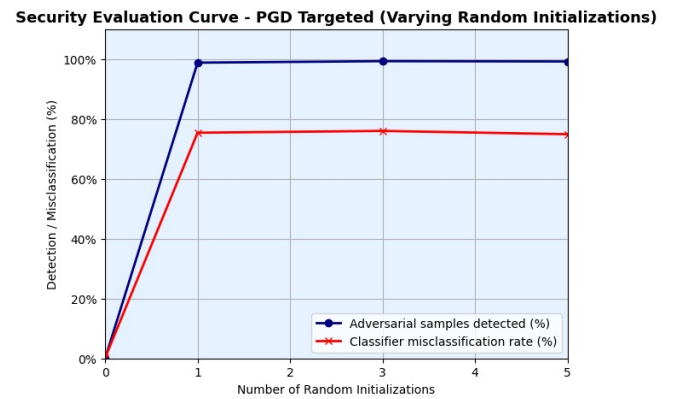
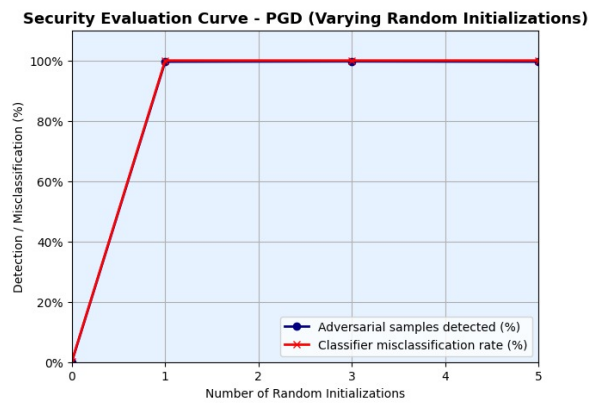
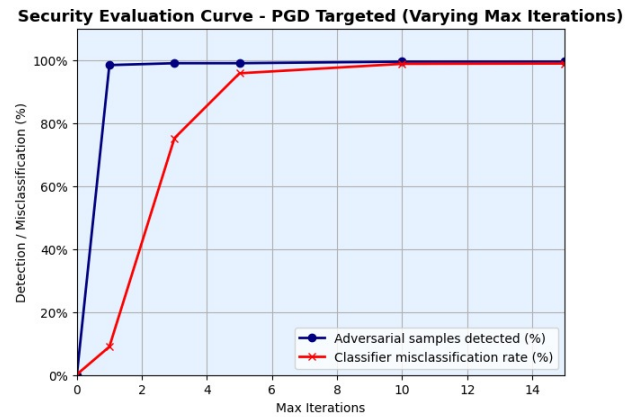
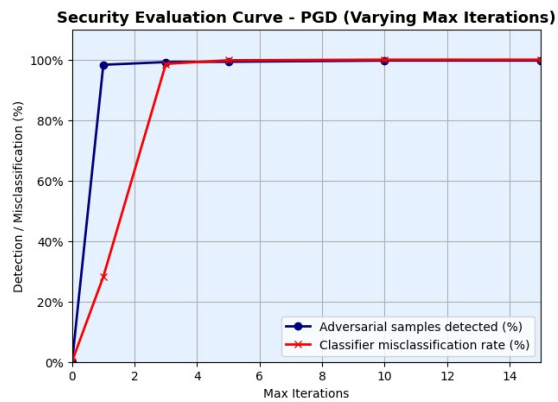
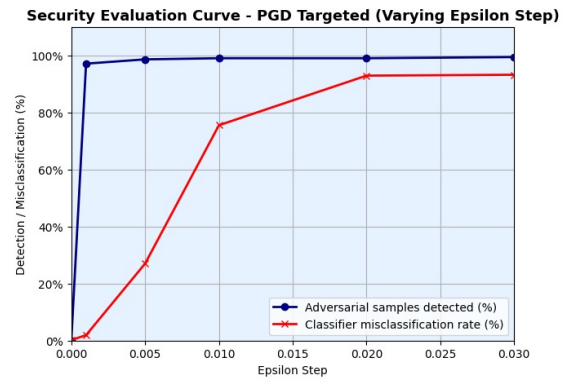
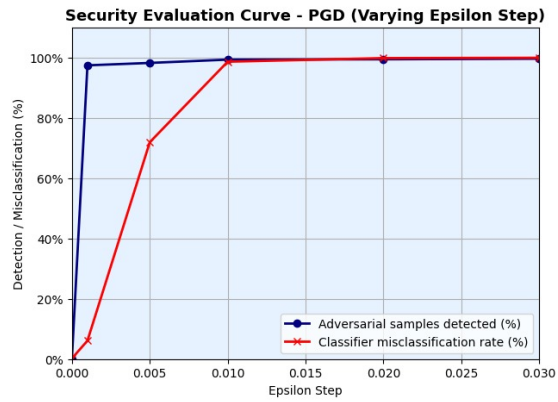
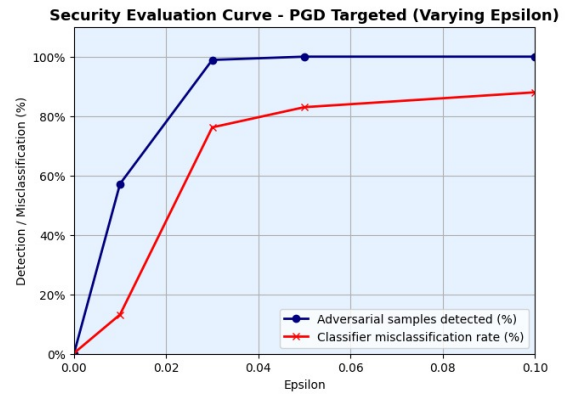
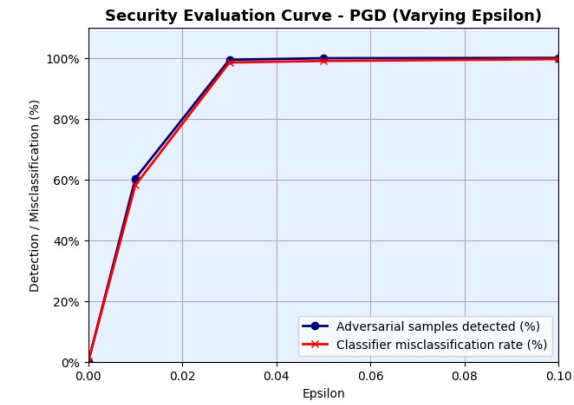


Figure 23: Security Evaluation Curves for PGD (error-generic variant) with Detector Accuracy.

Figure 24: Security Evaluation Curves for PGD (error-specific variant) with Detector Accuracy.

5.3 | Integration of Preprocessing and Detector

Having analyzed the effectiveness of the preprocessing pipeline and the adversarial detector separately, we then explored how these two mechanisms can work together to provide a more comprehensive defense. Although the detector was specifically trained on gradient-based attacks (FGSM, BIM, and PGD), we recognized that in practice it is impossible to know in advance which type of adversarial attack might be encountered. For this reason, we also tested the detector on adversarial examples generated with the Carlini & Wagner attack, using the single parameter configuration we had available due to computational constraints. In these tests, the detector correctly identified about 70% of the error-generic samples and about 42% of the error-specific samples.

Given these findings, our integration strategy is straightforward: incoming samples are first processed by the detector. Those that are flagged as adversarial are immediately discarded, while the remaining inputs (deemed clean) are passed through the preprocessing module. This second step is particularly important because the detector may not capture all adversarial examples, particularly if they were generated by different attack types or with parameters not seen during training. The preprocessing stage therefore acts as an additional line of defense, helping to mitigate any residual adversarial perturbations that might still affect the classifier.

Based on the combined results of our experiments, this integrated approach, using the detector as the first line of defense and the preprocessing module as a final safeguard, should provide NN1 with improved resilience against the adversarial attacks we evaluated.

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